

Experiment 'p_3_10_temperature_altitude' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_3_10_temperature_altitude

Natural language statement: null

Method signature: p_3_10_temperature_altitude (t: real, a: nat) returns (s: string)

Ensures

- if t < 0.0 then s == "solid" else if t == 0.0 then s == "solid or liquid" else if 0.0 < t < 100.0 - (a as real / 300.0) then s == "liquid" else if t == 100.0 - (a as real / 300.0) then s == "liquid or gaseous" else s == "gaseous"

Requires

- t >= -273.15 && a <= 10000

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_3_10_temperature_altitude (t: real, a: nat) returns (s: string)`

The method should respect the following contract:

requires `t >= -273.15 && a <= 10000`, ensures if `t < 0.0` then `s == "solid"` else if `t == 0.0` then `s == "solid or liquid"` else if `0.0 < t < 100.0 - (a as real / 300.0)` then `s == "liquid"` else if `t == 100.0 - (a as real / 300.0)` then `s == "liquid or gaseous"` else `s == "gaseous"`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn3EBwp5hNvdWuJvgVNU9ckyhNIOe

Model: gpt-5.5-2026-04-23

Created at: 1780582851

Finish reason: stop

Usage: 464 tokens in, and 659 tokens out

```
{
  var boiling: real := 100.0 - (a as real) / 300.0;
  if t < 0.0 {
    s := "solid";
  } else if t == 0.0 {
    s := "solid or liquid";
  } else if 0.0 < t && t < boiling {
    s := "liquid";
  } else if t == boiling {
    s := "liquid or gaseous";
  } else {
    s := "gaseous";
  }
}
```

Final Program

```

method p_3_10_temperature_altitude (t: real, a: nat) returns (s: string)
  requires t >= -273.15 && a <= 10000
  ensures if t < 0.0 then s == "solid" else if t == 0.0 then s == "solid or liquid"
    ↪ else if 0.0 < t < 100.0 - (a as real / 300.0) then s == "liquid" else if t
    ↪ == 100.0 - (a as real / 300.0) then s == "liquid or gaseous" else s == "
    ↪ gaseous"
{
  var boiling: real := 100.0 - (a as real) / 300.0;
  if t < 0.0 {
    s := "solid";
  } else if t == 0.0 {
    s := "solid or liquid";
  } else if 0.0 < t && t < boiling {
    s := "liquid";
  } else if t == boiling {
    s := "liquid or gaseous";
  } else {
    s := "gaseous";
  }
}

```

Total Token Usage

Input tokens: 464

Output tokens: 659

Reasoning tokens: 512

Sum of 'total tokens': 1123

Experiment Timings

Overall Experiment started at 1780582850863, ended at 1780582914126, lasting 63263ms (63.26 seconds)

Iteration #1 started at 1780582850921, ended at 1780582914126, lasting 63205ms (63.21 seconds)

Experiment 'p_3_1_classify_int' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_3_1_classify_int

Natural language statement: null

Method signature: p_3_1_classify_int (x: int) returns (s: string)

Ensures

- $x == 0 \implies s == \text{"zero"}$
- $x > 0 \implies s == \text{"positive"}$
- $x < 0 \implies s == \text{"negative"}$

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_3_1_classify_int (x: int) returns (s: string)`

The method should respect the following contract:

`ensures x == 0 ==> s == "zero", ensures x > 0 ==> s == "positive", ensures x < 0 ==> s ==`
 ↪ `"negative"`

Produce and show only the Dafny body of this method, including the curly braces that
 ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
 ↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn33zHRBl92XftOPpFZBGTCh4vv8x

Model: gpt-5.5-2026-04-23

Created at: 1780582219

Finish reason: stop

Usage: 300 tokens in, and 149 tokens out

```
{
  if x == 0 {
    s := "zero";
  } else if x > 0 {
    s := "positive";
  } else {
    s := "negative";
  }
}
```

Final Program

```
method p_3_1_classify_int (x: int) returns (s: string)
  ensures x == 0 ==> s == "zero"
  ensures x > 0 ==> s == "positive"
  ensures x < 0 ==> s == "negative"
{
  if x == 0 {
```

```
    s := "zero";  
  } else if x > 0 {  
    s := "positive";  
  } else {  
    s := "negative";  
  }  
}
```

Total Token Usage

Input tokens: 300

Output tokens: 149

Reasoning tokens: 79

Sum of 'total tokens': 449

Experiment Timings

Overall Experiment started at 1780582205146, ended at 1780582267137, lasting 61991ms (61.99 seconds)

Iteration #1 started at 1780582205399, ended at 1780582267135, lasting 61736ms (61.74 seconds)

Experiment 'p_3_2_classify_real' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_3_2_classify_real

Natural language statement: null

Method signature: p_3_2_classify_real (x: real) returns (s: string)

Ensures

- $x == 0.0 \implies s == \text{"zero, small"}$
- $1.0 > x > 0.0 \implies s == \text{"positive, small"}$
- $1.0 \leq x < 1000000.0 \implies s == \text{"positive"}$
- $1000000.0 \leq x \implies s == \text{"positive, large"}$
- $0.0 > x > -1.0 \implies s == \text{"negative, small"}$
- $-1.0 \geq x > -1000000.0 \implies s == \text{"negative"}$
- $-1000000.0 \geq x \implies s == \text{"negative, large"}$

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_3_2_classify_real (x: real) returns (s: string)`

The method should respect the following contract:

```
ensures x == 0.0 ==> s == "zero, small", ensures 1.0 > x > 0.0 ==> s == "positive, small  
  ↳ ", ensures 1.0 <= x < 1000000.0 ==> s == "positive", ensures 1000000.0 <= x ==> s  
  ↳ == "positive, large", ensures 0.0 > x > -1.0 ==> s == "negative, small", ensures  
  ↳ -1.0 >= x > -1000000.0 ==> s == "negative", ensures -1000000.0 >= x ==> s == "  
  ↳ negative, large"
```

Produce and show only the Dafny body of this method, including the curly braces that
 ↳ surround it. Do not show the signature nor contract. You must not use 'assume' nor
 ↳ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn34qglfPvbEfiHyOxJa7OVs2vhMG

Model: gpt-5.5-2026-04-23

Created at: 1780582272

Finish reason: stop

Usage: 494 tokens in, and 603 tokens out

```
{  
  if x == 0.0 {  
    s := "zero, small";  
  } else if x > 0.0 {  
    if x < 1.0 {  
      s := "positive, small";  
    } else if x < 1000000.0 {  
      s := "positive";  
    } else {  
      s := "positive, large";  
    }  
  } else {  
    if x > -1.0 {  
      s := "negative, small";  
    } else if x > -1000000.0 {  
      s := "negative";  
    } else {
```

```
        s := "negative, large";
    }
}
}
```

Final Program

```
method p_3_2_classify_real (x: real) returns (s: string)
  ensures x == 0.0 ==> s == "zero, small"
  ensures 1.0 > x > 0.0 ==> s == "positive, small"
  ensures 1.0 <= x < 1000000.0 ==> s == "positive"
  ensures 1000000.0 <= x ==> s == "positive, large"
  ensures 0.0 > x > -1.0 ==> s == "negative, small"
  ensures -1.0 >= x > -1000000.0 ==> s == "negative"
  ensures -1000000.0 >= x ==> s == "negative, large"
{
  if x == 0.0 {
    s := "zero, small";
  } else if x > 0.0 {
    if x < 1.0 {
      s := "positive, small";
    } else if x < 1000000.0 {
      s := "positive";
    } else {
      s := "positive, large";
    }
  } else {
    if x > -1.0 {
      s := "negative, small";
    } else if x > -1000000.0 {
      s := "negative";
    } else {
      s := "negative, large";
    }
  }
}
```

Total Token Usage

Input tokens: 494

Output tokens: 603

Reasoning tokens: 423

Sum of 'total tokens': 1097

Experiment Timings

Overall Experiment started at 1780582270650, ended at 1780582320074, lasting 49424ms (49.42 seconds)

Iteration #1 started at 1780582270660, ended at 1780582320071, lasting 49411ms (49.41 seconds)

Experiment ‘p_3_3_number_of_digits’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 2

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 2

Problem Specification

Problem name: p_3_3_number_of_digits

Natural language statement: null

Method signature: p_3_3_number_of_digits (i: int) returns (n: nat)

Ensures

- $i \geq 0 \implies n == \text{digits}(i)$
- $i < 0 \implies n == \text{digits}(-i)$

Requires

- $10000000000 > i > -10000000000$

Functional Code Given

```
function digits (n: nat): nat
{
  if 0 <= n <= 9 then 1
  else 1 + digits (n / 10)
}
```

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_3_3_number_of_digits (i: int) returns (n: nat)`

The method should respect the following contract:

`requires 10000000000 > i > -10000000000, ensures i >= 0 ==> n == digits(i), ensures i < 0
 ↪ ==> n == digits(-i)`

The contract uses the following dafny code:

```
function digits (n: nat): nat
{
  if 0 <= n <= 9 then 1
  else 1 + digits (n / 10)
}
```

These function(s) must not be used in your implementation of the method.

Produce and show only the Dafny body of this method, including the curly braces that
 ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
 ↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn35i1THYrYDrTz0LVIVD3h6CD7Bg

Model: gpt-5.5-2026-04-23

Created at: 1780582326

Finish reason: stop

Usage: 428 tokens in, and 747 tokens out

```
{
  var x: int;
  if i < 0 {
    x := -i;
  } else {
    x := i;
  }

  if x < 10 {
    n := 1;
```

```

} else if x < 100 {
  n := 2;
} else if x < 1000 {
  n := 3;
} else if x < 10000 {
  n := 4;
} else if x < 100000 {
  n := 5;
} else if x < 1000000 {
  n := 6;
} else if x < 10000000 {
  n := 7;
} else if x < 100000000 {
  n := 8;
} else if x < 1000000000 {
  n := 9;
} else {
  n := 10;
}
}

```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

```

problem_data/p_3_3_number_of_digits/program.dfy(11,0): Error: a postcondition could not
  ↳ be proved on this return path

```

```

|
11 | {
| ^

```

```

problem_data/p_3_3_number_of_digits/program.dfy(9,22): Related location: this is the
  ↳ postcondition that could not be proved

```

```

|
9 | ensures i >= 0 ==> n == digits(i)
| ^^

```

```

problem_data/p_3_3_number_of_digits/program.dfy(11,0): Error: a postcondition could not
  ↳ be proved on this return path

```

```

|
11 | {
| ^

```

```

problem_data/p_3_3_number_of_digits/program.dfy(10,21): Related location: this is the
  ↳ postcondition that could not be proved

```

```

|
10 | ensures i < 0 ==> n == digits(-i)
| ^^

```

Dafny program verifier finished with 2 verified, 2 errors

Correct the error in code by altering only the method body. Produce and show only the
→ Dafny body, including the curly braces that surround it. Do not show the signature
→ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn36foJ9tz876uteDKk6XoBLpMoHf

Model: gpt-5.5-2026-04-23

Created at: 1780582385

Finish reason: stop

Usage: 1020 tokens in, and 5033 tokens out

```
{
  var x: nat;
  if i < 0 {
    assert -i > 0;
    x := -i;
  } else {
    assert i >= 0;
    x := i;
  }

  var y: nat := x;
  n := 1;
  while y >= 10
    invariant n + digits(y) == 1 + digits(x)
    decreases y
  {
    var z: nat := y / 10;
    assert z < y;
    assert !(0 <= y <= 9);
    assert digits(y) == 1 + digits(z);
    calc {
      (n + 1) + digits(z);
      ==
      n + (1 + digits(z));
      ==
      n + digits(y);
      ==
      1 + digits(x);
    }
    n := n + 1;
    y := z;
  }

  assert 0 <= y <= 9;
  assert digits(y) == 1;
  assert n + digits(y) == 1 + digits(x);
  assert n + 1 == 1 + digits(x);
}
```

```

assert 1 + digits(x) == digits(x) + 1;
assert n + 1 == digits(x) + 1;
assert n == digits(x);

if i >= 0 {
  assert x == i;
  assert n == digits(i);
} else {
  assert x == -i;
  assert n == digits(-i);
}
}

```

Final Program

```

function digits (n: nat): nat
{
  if 0 <= n <= 9 then 1
  else 1 + digits (n / 10)
}

method p_3_3_number_of_digits (i: int) returns (n: nat)
  requires 10000000000 > i > -10000000000
  ensures i >= 0 ==> n == digits(i)
  ensures i < 0 ==> n == digits(-i)
{
  var x: nat;
  if i < 0 {
    assert -i > 0;
    x := -i;
  } else {
    assert i >= 0;
    x := i;
  }

  var y: nat := x;
  n := 1;
  while y >= 10
    invariant n + digits(y) == 1 + digits(x)
    decreases y
  {
    var z: nat := y / 10;
    assert z < y;
    assert !(0 <= y <= 9);
    assert digits(y) == 1 + digits(z);
    calc {
      (n + 1) + digits(z);
      ==
      n + (1 + digits(z));
      ==
    }
  }
}

```

```

    n + digits(y);
    ==
    1 + digits(x);
}
n := n + 1;
y := z;
}

assert 0 <= y <= 9;
assert digits(y) == 1;
assert n + digits(y) == 1 + digits(x);
assert n + 1 == 1 + digits(x);
assert 1 + digits(x) == digits(x) + 1;
assert n + 1 == digits(x) + 1;
assert n == digits(x);

if i >= 0 {
    assert x == i;
    assert n == digits(i);
} else {
    assert x == -i;
    assert n == digits(-i);
}
}

```

Total Token Usage

Input tokens: 1448

Output tokens: 5780

Reasoning tokens: 5120

Sum of 'total tokens': 7228

Experiment Timings

Overall Experiment started at 1780582323650, ended at 1780582555663, lasting 232013ms (232.01 seconds)

Iteration #1 started at 1780582323677, ended at 1780582384977, lasting 61300ms (61.30 seconds)

Iteration #2 started at 1780582384988, ended at 1780582555663, lasting 170675ms (170.68 seconds)

Experiment ‘p_3_4_classification’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_3_4_classification

Natural language statement: null

Method signature: p_3_4_classification (i: int, j: int, k: int) returns (s: string)

Ensures

- if i == j == k then s == "all the same" else if i != j && j != k && i != k then s == "all different" else s == "neither"

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_3_4_classification (i: int, j: int, k: int) returns (s: string)`

The method should respect the following contract:

`ensures if i == j == k then s == "all the same" else if i != j && j != k && i != k then s
 ↪ == "all different" else s == "neither"`

Produce and show only the Dafny body of this method, including the curly braces that
 ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
 ↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn39UqXhhBzHnbmHdNUf8iXLv8dxC

Model: gpt-5.5-2026-04-23

Created at: 1780582560

Finish reason: stop

Usage: 332 tokens in, and 200 tokens out

```
{  
  if i == j && j == k {  
    s := "all the same";  
  } else if i != j && j != k && i != k {  
    s := "all different";  
  } else {  
    s := "neither";  
  }  
}
```

Final Program

```
method p_3_4_classification (i: int, j: int, k: int) returns (s: string)  
  ensures if i == j == k then s == "all the same" else if i != j && j != k && i != k  
    ↪ then s == "all different" else s == "neither"  
{  
  if i == j && j == k {  
    s := "all the same";  
  }
```

```
} else if i != j && j != k && i != k {  
  s := "all different";  
} else {  
  s := "neither";  
}  
}
```

Total Token Usage

Input tokens: 332

Output tokens: 200

Reasoning tokens: 116

Sum of 'total tokens': 532

Experiment Timings

Overall Experiment started at 1780582559727, ended at 1780582595512, lasting 35785ms (35.79 seconds)

Iteration #1 started at 1780582559729, ended at 1780582595512, lasting 35783ms (35.78 seconds)

Experiment 'p_3_5_order' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_3_5_order

Natural language statement: null

Method signature: p_3_5_order (i: int, j: int, k: int) returns (s: string)

Ensures

- if $i < j < k$ then $s == \text{"increasing"}$ else if $i > j \ \&\& \ j > k$ then $s == \text{"decreasing"}$ else $s == \text{"neither"}$

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_3_5_order (i: int, j: int, k: int) returns (s: string)`

The method should respect the following contract:

`ensures if i < j < k then s == "increasing" else if i > j && j > k then s == "decreasing"`
 ↪ `else s == "neither"`

Produce and show only the Dafny body of this method, including the curly braces that
 ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
 ↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn3A9vhWRJPb4tpch1JghSzD7mhgd

Model: gpt-5.5-2026-04-23

Created at: 1780582601

Finish reason: stop

Usage: 320 tokens in, and 193 tokens out

```
{
  if i < j && j < k {
    s := "increasing";
  } else if i > j && j > k {
    s := "decreasing";
  } else {
    s := "neither";
  }
}
```

Final Program

```
method p_3_5_order (i: int, j: int, k: int) returns (s: string)
  ensures if i < j < k then s == "increasing" else if i > j && j > k then s == "
    ↪ decreasing" else s == "neither"
{
  if i < j && j < k {
    s := "increasing";
```

```
} else if i > j && j > k {  
  s := "decreasing";  
} else {  
  s := "neither";  
}  
}
```

Total Token Usage

Input tokens: 320

Output tokens: 193

Reasoning tokens: 114

Sum of 'total tokens': 513

Experiment Timings

Overall Experiment started at 1780582600827, ended at 1780582671567, lasting 70740ms (70.74 seconds)

Iteration #1 started at 1780582600830, ended at 1780582671567, lasting 70737ms (70.74 seconds)

Experiment ‘p_3_6_parametrised_order’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_3_6_parametrised_order

Natural language statement: null

Method signature: p_3_6_parametrised_order (i: int, j: int, k: int, strict: bool) returns (s: string)

Ensures

- if strict then if i < j < k then s == "increasing" else if i > j && j > k then s == "decreasing" else s == "neither" else if i <= j <= k then if i == j == k then s == "increasing and decreasing" else s == "increasing" else if i >= j && j >= k then s == "decreasing" else s == "neither"

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_3_6_parametrised_order (i: int, j: int, k: int, strict: bool ) returns (s: string)`

The method should respect the following contract:

```
ensures if strict then if i < j < k then s == "increasing" else if i > j && j > k then s
  ↳ == "decreasing" else s == "neither" else if i <= j <= k then if i == j == k then s
  ↳ == "increasing and decreasing" else s == "increasing" else if i >= j && j >= k
  ↳ then s == "decreasing" else s == "neither"
```

Produce and show only the Dafny body of this method, including the curly braces that
↳ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↳ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn3BYeiplW4zGzORLFaehNFG4p1F

Model: gpt-5.5-2026-04-23

Created at: 1780582688

Finish reason: stop

Usage: 446 tokens in, and 293 tokens out

```
{
  if strict {
    if i < j < k {
      s := "increasing";
    } else if i > j && j > k {
      s := "decreasing";
    } else {
      s := "neither";
    }
  } else {
    if i <= j <= k {
      if i == j == k {
        s := "increasing and decreasing";
      } else {
        s := "increasing";
      }
    } else if i >= j && j >= k {
      s := "decreasing";
    }
  }
}
```

```

    } else {
      s := "neither";
    }
  }
}

```

Final Program

```

method p_3_6_parametrised_order (i: int, j: int, k: int, strict: bool ) returns (s:
  ↪ string)
  ensures if strict then if i < j < k then s == "increasing" else if i > j && j > k
    ↪ then s == "decreasing" else s == "neither" else if i <= j <= k then if i ==
    ↪ j == k then s == "increasing and decreasing" else s == "increasing" else
    ↪ if i >= j && j >= k then s == "decreasing" else s == "neither"
{
  if strict {
    if i < j < k {
      s := "increasing";
    } else if i > j && j > k {
      s := "decreasing";
    } else {
      s := "neither";
    }
  } else {
    if i <= j <= k {
      if i == j == k {
        s := "increasing and decreasing";
      } else {
        s := "increasing";
      }
    } else if i >= j && j >= k {
      s := "decreasing";
    } else {
      s := "neither";
    }
  }
}

```

Total Token Usage

Input tokens: 446

Output tokens: 293

Reasoning tokens: 118

Sum of 'total tokens': 739

Experiment Timings

Overall Experiment started at 1780582686668, ended at 1780582723347, lasting 36679ms (36.68 seconds)

Iteration #1 started at 1780582686669, ended at 1780582723347, lasting 36678ms (36.68 seconds)

Experiment 'p_3_7_order' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_3_7_order

Natural language statement: null

Method signature: p_3_7_order (i: int, j: int, k: int) returns (s: string)

Ensures

- `if (i < j < k) || (i > j > k) then s == "in order" else s == "not in order"`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_3_7_order (i: int, j: int, k: int) returns (s: string)`

The method should respect the following contract:

`ensures if (i < j < k) || (i > j > k) then s == "in order" else s == "not in order"`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn3CAcFktC1z0P6YXa9o7wcjBVO1F

Model: gpt-5.5-2026-04-23

Created at: 1780582726

Finish reason: stop

Usage: 310 tokens in, and 212 tokens out

```
{
  if (i < j && j < k) || (i > j && j > k) {
    s := "in order";
  } else {
    s := "not in order";
  }
}
```

Final Program

```
method p_3_7_order (i: int, j: int, k: int) returns (s: string)
  ensures if (i < j < k) || (i > j > k) then s == "in order" else s == "not in order"
  → "
{
  if (i < j && j < k) || (i > j && j > k) {
    s := "in order";
  } else {
    s := "not in order";
  }
}
```

}

Total Token Usage

Input tokens: 310

Output tokens: 212

Reasoning tokens: 142

Sum of 'total tokens': 522

Experiment Timings

Overall Experiment started at 1780582725622, ended at 1780582763603, lasting 37981ms (37.98 seconds)

Iteration #1 started at 1780582725623, ended at 1780582763600, lasting 37977ms (37.98 seconds)

Experiment 'p_3_8_two_pairs' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_3_8_two_pairs

Natural language statement: null

Method signature: p_3_8_two_pairs (i: int, j: int, k: int, l: int) returns (s: string)

Ensures

- if (i == j && k == l) || (i == k && j == l) || (i == l && j == k) then s == "two pairs"
else s == "not two pairs"

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_3_8_two_pairs (i: int, j: int, k: int, l: int) returns (s: string)`

The method should respect the following contract:

`ensures if (i == j && k == l) || (i == k && j == l) || (i == l && j == k) then s == "two
↪ pairs" else s == "not two pairs"`

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn3CoNtQrRuwyUPGEZcuwucD2nXCT

Model: gpt-5.5-2026-04-23

Created at: 1780582766

Finish reason: stop

Usage: 348 tokens in, and 164 tokens out

```
{  
  if (i == j && k == l) || (i == k && j == l) || (i == l && j == k) {  
    s := "two pairs";  
  } else {  
    s := "not two pairs";  
  }  
}
```

Final Program

```
method p_3_8_two_pairs (i: int, j: int, k: int, l: int) returns (s: string)  
  ensures if (i == j && k == l) || (i == k && j == l) || (i == l && j == k) then s  
    ↪ == "two pairs" else s == "not two pairs"  
{  
  if (i == j && k == l) || (i == k && j == l) || (i == l && j == k) {  
    s := "two pairs";  
  } else {  
    s := "not two pairs";  
  }  
}
```

```
}  
}
```

Total Token Usage

Input tokens: 348

Output tokens: 164

Reasoning tokens: 84

Sum of ‘total tokens’: 512

Experiment Timings

Overall Experiment started at 1780582766455, ended at 1780582811043, lasting 44588ms (44.59 seconds)

Iteration #1 started at 1780582766456, ended at 1780582811043, lasting 44587ms (44.59 seconds)

Experiment 'p_3_9_temperature' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_3_9_temperature

Natural language statement: null

Method signature: p_3_9_temperature (t: real, c: char) returns (s: string)

Ensures

- if c == 'C' then if t < 0.0 then s == "solid" else if t == 0.0 then s == "solid or liquid" else if 0.0 < t < 100.0 then s == "liquid" else if t == 100.0 then s == "liquid or gaseous" else s == "gaseous" else if t < 32.0 then s == "solid" else if t == 32.0 then s == "solid or liquid" else if 32.0 < t < 212.0 then s == "liquid" else if t == 212.0 then s == "liquid or gaseous" else s == "gaseous"

Requires

- (c == 'C' && t >= -273.15) || (c == 'F' && t >= -459.67)
- c == 'C' || c == 'F'

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_3_9_temperature (t: real, c: char) returns (s: string)`

The method should respect the following contract:

```
requires (c == 'C' && t >= -273.15) || (c == 'F' && t >= -459.67), requires c == 'C' || c
  ↪ == 'F', ensures if c == 'C' then if t < 0.0 then s == "solid" else if t == 0.0
  ↪ then s == "solid or liquid" else if 0.0 < t < 100.0 then s == "liquid" else if t
  ↪ == 100.0 then s == "liquid or gaseous" else s == "gaseous" else if t < 32.0 then s
  ↪ == "solid" else if t == 32.0 then s == "solid or liquid" else if 32.0 < t < 212.0
  ↪ then s == "liquid" else if t == 212.0 then s == "liquid or gaseous" else s == "
  ↪ gaseous"
```

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn3DbBoqoKfKrfogHSUGg1IElaURQ

Model: gpt-5.5-2026-04-23

Created at: 1780582815

Finish reason: stop

Usage: 638 tokens in, and 489 tokens out

```
{
  if c == 'C' {
    if t < 0.0 {
      s := "solid";
    } else if t == 0.0 {
      s := "solid or liquid";
    } else if t < 100.0 {
      s := "liquid";
    } else if t == 100.0 {
      s := "liquid or gaseous";
    } else {
      s := "gaseous";
    }
  } else {
    if t < 32.0 {
```

```

    s := "solid";
  } else if t == 32.0 {
    s := "solid or liquid";
  } else if t < 212.0 {
    s := "liquid";
  } else if t == 212.0 {
    s := "liquid or gaseous";
  } else {
    s := "gaseous";
  }
}
}

```

Final Program

```

method p_3_9_temperature (t: real, c: char) returns (s: string)
  requires (c == 'C' && t >= -273.15) || (c == 'F' && t >= -459.67)
  requires c == 'C' || c == 'F'
  ensures if c == 'C' then if t < 0.0 then s == "solid" else if t == 0.0 then s == "
    ↪ solid or liquid" else if 0.0 < t < 100.0 then s == "liquid" else if t ==
    ↪ 100.0 then s == "liquid or gaseous" else s == "gaseous" else if t < 32.0
    ↪ then s == "solid" else if t == 32.0 then s == "solid or liquid" else if
    ↪ 32.0 < t < 212.0 then s == "liquid" else if t == 212.0 then s == "liquid or
    ↪ gaseous" else s == "gaseous"
{
  if c == 'C' {
    if t < 0.0 {
      s := "solid";
    } else if t == 0.0 {
      s := "solid or liquid";
    } else if t < 100.0 {
      s := "liquid";
    } else if t == 100.0 {
      s := "liquid or gaseous";
    } else {
      s := "gaseous";
    }
  } else {
    if t < 32.0 {
      s := "solid";
    } else if t == 32.0 {
      s := "solid or liquid";
    } else if t < 212.0 {
      s := "liquid";
    } else if t == 212.0 {
      s := "liquid or gaseous";
    } else {
      s := "gaseous";
    }
  }
}

```

}

Total Token Usage

Input tokens: 638

Output tokens: 489

Reasoning tokens: 246

Sum of 'total tokens': 1127

Experiment Timings

Overall Experiment started at 1780582814552, ended at 1780582845691, lasting 31139ms (31.14 seconds)

Iteration #1 started at 1780582814566, ended at 1780582845691, lasting 31125ms (31.13 seconds)

